Shaon Kanti Dhar

Project Leader | *Samsung R&D Institute Bangladesh*
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Portfolio: https://shawonnstu.github.io/ · Dhaka, Bangladesh

RESEARCH INTERESTS

Security and Privacy of Connected Devices (IoT, AR/VR and wearables), AI Agents and LLM Systems, Speech and Audio Processing, and Edge AI (exploring speaker verification, context-aware authentication, wearable sensing and retrieval-augmented generation)

EDUCATIONAL BACKGROUND

Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh
B.Sc. in Information and Communication Engineering **2015 – 2019**

- CGPA: 3.51/4.00
- Final-year project: Travel Assistant, an Android app with automated image screening

Feni Government College, Feni, Bangladesh
Higher Secondary Certificate (HSC), Science **2014**

- GPA: 4.80/5.00

Feni Government Pilot High School, Feni, Bangladesh
Secondary School Certificate (SSC), Science **2012**

- GPA: 5.00/5.00

RESEARCH EXPERIENCE

Flicker Reduction in Multi-Camera Video Recording **Aug 2025**
Samsung Electronics* · *Co-inventor **Patent**

- Method that synchronizes camera settings to reduce light flicker when several cameras record at once

Missed-Speech Recovery for Voice Calls **Mar 2025**
Samsung R&D Institute Bangladesh* · *Co-inventor **Patent research**

- Detects when parts of a call are lost to disconnection, poor audio or noise, and recovers the missed speech so the conversation continues without gaps

Security for Bluetooth Multipoint Earbuds **Dec 2024**
Samsung R&D Institute Bangladesh* · *Co-inventor **Invention proposal**

- Identified how shared or multipoint earbuds can leak calls and private notifications to the wrong listener
- Proposed per-earbud trust levels from on-device speaker verification and contextual signals (wear sensors, connection state, location)
- Designed an audio sensitivity classifier using rules plus ASR and NLP; audio is routed only to sufficiently trusted earbuds

Shoulder-Surfing Defence for VR Typing **May 2024 – Aug 2024**
Samsung R&D Institute Bangladesh* · *Co-inventor **Invention proposal**

- Proposed a virtual keyboard that moves, tilts or resizes every few keystrokes within the user's comfortable reach, so filmed hand movements cannot reveal a password
- Added a security mode that adapts to the user's surroundings

Context-Aware Authentication Against Fraud **Feb 2024 – Apr 2024**
Samsung R&D Institute Bangladesh* · *Co-inventor **Patent research**

- Authentication framework that senses behaviour and context, uses an LLM to spot deceptive or coerced situations, and raises security step by step to block fraudulent authorization

Conversational Q&A on Audio Media **Jan 2024**
Samsung R&D Institute Bangladesh* · *Co-inventor

- Built a question-answering framework for audio embedded in documents and web pages: extracts the audio, transcribes it with Whisper and answers with GPT-3.5
- Makes meetings, lectures and podcasts searchable by asking

Overexertion Warning for Wearables **Dec 2023**
Samsung R&D Institute Bangladesh · Co-inventor **Invention proposal**

- Proposed a smartwatch feature that warns users before they over-exercise, estimating blood lactate with optical sensing instead of bulky sweat pads
- Combines heart rate, temperature, motion, sleep and food intake to avoid false alarms

Accidental-Touch Filtering for Wireless Earbuds **Feb 2022 – Mar 2022**
Samsung Electronics · Co-inventor **Patent**

- Method using sensor data to separate intentional touches from accidental ones on earbud touchpads
- **Patent Link:** US20230229260A1

AI PROJECTS

PQM Agent (Process Quality Management) **Dec 2025 – Present**
Samsung R&D Institute Bangladesh · Designer & Developer

- Runs automatically on every pushed commit: static analysis, unit tests, software architecture maturity (SAM) scoring and a regression check
- AI-driven SQE solution that saves 30% of release time; used across the whole organization

Enterprise AI Assistant **Jul 2025 – Mar 2026**
Enterprise AI platform · Architect & Lead Developer

- Architected an agentic RAG assistant that answers questions from documents and data, with page-level citations; used by 500+ engineers
- Hybrid retrieval and reranking on open-weight LLMs; role-based and row-level access control

AI Test Management **Dec 2023 – Jun 2024**
Samsung R&D Institute Bangladesh · Developer

- Fine-tuned a model on the application UI/UX guide to generate QA test cases
- Built the LLM-based system that versions test cases through the full release cycle; used by 40+ QA engineers
- Automated test procedure applied successfully in 60+ projects across the organization

SmartThings Pro B2B AI Agent **2025 – Present**
Samsung R&D Institute Bangladesh · Architect

- Architected the AI agent for Samsung's B2B building-management platform and led its rollout to Starbucks

EXPERIENCE

Project Leader **Dec 2025 – Present**
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh · SmartThings Pro

- Lead a team of 12 engineers across frontend, backend, database and cloud services
- Designed and built the PQM Agent, an AI-driven SQE solution that checks every pushed commit and saves 30% of release time; used across the whole organization
- Architected the SmartThings Pro B2B AI Agent and rolled it out to Starbucks
- Achieved a 15% improvement in load time and rendering speed on the SmartThings Pro platform through request parallelization, client-side caching, query orchestration, list virtualization and real-time update processing
- Set up an AI-assisted development environment and internal workflows using LLMs, RAG and tool integration

Technical skills: Technical project leadership, Vue.js, Front-end development, Backend development, Spring Boot

Lead Engineer **Mar 2025 – Dec 2025**
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh · Lounge, Virtual Studio, SmartThings Pro

- Led 2 SmartThings retail web apps, Lounge and Virtual Studio, that let visitors control SmartThings devices at Samsung showrooms and trade shows
- Architected and led the development of an enterprise agentic RAG assistant over documents and structured data, used by 500+ engineers
- Owned application architecture, API integration and real-time device interaction
- Built SmartThings Pro's Map View, multi-site management and property dashboard

Technical skills: Agentic RAG, Three.js, Unity, Next.js, Kotlin, Micronaut

Senior Software Engineer **Mar 2022 – Mar 2025**
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh · AI Test Management

- Built an AI test-management system: fine-tuned a model on our UI/UX guide to generate QA test cases, and built the LLM-based system that versions them through the full release cycle; used by 40+ QA engineers
- Applied the automated test procedure successfully in 60+ projects across the organization

Technical skills: LLM fine-tuning, AI test automation, Test-case generation, Java, React.js, Spring Boot, MySQL

Software Engineer **Oct 2019 – Mar 2022**
Samsung R&D Institute Bangladesh, Dhaka, Bangladesh · 5G Network Data Visualization

- Built the React web apps and Spring Boot REST APIs that turn Samsung 5G network data into interactive views engineers can explore

Technical skills: Java, Spring Boot, React, MySQL, MongoDB, Microservices

Technical Content Writer **Feb 2016 – Oct 2019**
NSTU Code Academy (competitive programming blog)

- Wrote problem solutions and plain-language algorithm explanations: Blog

Software Engineering Intern **Aug 2018**
Opus Technology Limited, Dhaka, Bangladesh

- Designed and built the base of an Android app with Java and Firebase Realtime Database

ACHIEVEMENTS

- **Excellence Award**, Samsung R&D Institute Bangladesh (Jan – Mar 2025) – for research work on missed-speech recovery for voice calls and Bluetooth multipoint security
- **Excellence Award**, Samsung R&D Institute Bangladesh (Oct – Dec 2024) – for work on Next SmartThings and Lounge, showcased at CES 2025, Las Vegas
- **Icon Award**, Samsung R&D Institute Bangladesh (May – Jun 2024) – for work on Virtual Studio, the SmartThings retail showroom for places with no devices and weak networks
- **Excellence Award**, Samsung R&D Institute Bangladesh (Jan – Mar 2024) – for work on AI Test Management, a fine-tuned model and system that generate and version QA test cases
- **Icon of the Month**, Samsung R&D Institute Bangladesh (Mar – Apr 2022) – for building major features of an internal tool used by Samsung teams
- **Excellence Award**, Samsung R&D Institute Bangladesh (Apr – Jun 2020) – for work on 5G Network Data Visualization, the web apps and APIs that present Samsung 5G network data
- 1st Runner-up, Intra University Programming Contest 2016, NSTU
- 2nd Runner-up, Intra ICE Programming Contest 2016, NSTU

LEADERSHIP AND SERVICE

- Vice President, ICE Programming Club, NSTU; mentored junior programmers
- Judge and problem setter: Intra ICE Programming Contest 2017, ICE FEST Programming Contest 2018, NSTU
- Problem setter: Intra University Programming Contest 2018 and 2019, NSTU

CERTIFICATIONS

- Samsung Global Software Certificate, Professional Level, Samsung Electronics (Feb 2022)
- Samsung Global Software Certificate, Advanced Level, Samsung Electronics (Oct 2019)

SKILLS

- **AI and ML:** LLM applications, agent workflows, LangChain, RAG, hybrid search and reranking, vLLM, text-to-SQL, MCP / tool integration
- **Languages:** TypeScript, Java, Kotlin, Python
- **Frontend:** Vue.js, React, Next.js, Three.js, Unity 3D
- **Backend:** Spring Boot, FastAPI, Micronaut, Spring Data / JPA, Microservices, SSE, OAuth, Event-driven systems
- **Data and Infrastructure:** PostgreSQL, pgvector, MySQL, MongoDB, Amazon Aurora, Redis, RabbitMQ, Elasticsearch, Docker, Nginx, Jenkins, AWS